

RetailPulse AI Sales Intelligence Platform

Comprehensive Internship Project Report & Documentation

1. Executive Summary

Overview: RetailPulse AI is a comprehensive, production-grade retail analytics platform engineered to empower businesses with actionable sales insights, demand forecasts, customer segmentation, churn prediction, and inventory health optimization. Built using Python, Pandas, Plotly, and Streamlit, the platform processes millions of historical transactional data points to deliver a multi-page interactive web application.

2. Project Objectives

- **Demand Forecasting:** Predicting future sales volumes using time-series modeling to ensure optimized supply chain planning.
- **Customer Intelligence:** Gauging customer behavior through Recency, Frequency, and Monetary (RFM) segmentation models.
- **Churn Prevention:** Identifying high-risk accounts based on inactivity thresholds to prevent customer churn.
- **Inventory Optimization:** Balancing stock levels using ABC inventory categorization and economic order quantity principles.

3. Technical Methodology

- **Phase 1 - Data Ingestion & ETL:** Ingested historical retail datasets using Pandas, handling missing values, filtering invalid prices, and removing canceled transaction records.
- **Phase 2 - Feature Engineering:** Engineered custom behavioral features including customer recency, order frequency, total monetary spend, and monthly time-series aggregations.
- **Phase 3 - Application Deployment:** Developed modular analytical components across 15 specialized views integrated seamlessly into a unified Streamlit application dashboard.

4. Key Performance Summary Table

The table below outlines the core metrics extracted from the global transactional dataset:

Metric Category	Indicator Value	Analytical Focus
Total Revenue	10.27M	Overall sales financial output
Total Orders	21K	Transactional volume throughput
Total Customers	4,313	Unique active buyer accounts
Global Reach	40 Countries	International market presence
Average Order Value	490.28	Mean basket value per transaction

5. Technology Stack

- **Programming Language:** Python 3.12 for core data processing and algorithmic computation.
- **Dashboard Framework:** Streamlit framework with custom CSS styling and native multi-page routing.
- **Visualization Engine:** Plotly Express for dynamic, interactive data visualizations and charts.
- **Deployment & Version Control:** Git and GitHub for robust version control and deployment integration.

6. Conclusion

Summary: RetailPulse AI successfully bridges the gap between raw data science modeling and practical business intelligence. By combining robust multi-page analytics, real-time filtering, and professional UI layout, the platform delivers a scalable solution for modern retail environments.